

The purpose of this assignment is to get experience with *exponential random graph modelling* (ERGM) of network structure, both in *thinking* and in *practise*. You are asked to reason about network mechanisms and assess the validity of your reasoning based on ERGM analyses and ERGM simulations. The *personal network data set* mentioned in the exercises below is distributed together with this assignment.

Idea behind the assignment

Developmental psychologists have found out that when children enter adolescence, two types of social standing in their peer group get differentiated (Mayeux, Houser & Dyches, 2011). On the one hand, there is **acceptance** in the sense of an individual being liked by many. On the other hand, there is **popularity** in the sense of a shared recognition that an individual has achieved high visibility and prestige in the peer group. Popular students in this sense are not necessarily accepted or liked, they even can display anti-social traits to bolster their social standing. Vice versa, accepted students in the above sense need not be recognized as popular, as they may be *just too nice* for winning a status competition.

Let us now add to this the sociometric element. The two constructs in boldface in the paragraph above typically get measured by so-called *peer nominations* in school class settings, which essentially are network questions, respectively, “*Whom in class do you like?*” and “*Who in class is popular?*” The acceptance and popularity of individuals is then calculated as their *indegree* in the two networks. In network research, by the way, the indegree often also gets called ‘popularity’ in a generic sense, which occasionally causes confusion in the literature.

Arguably, **friendship** may contain a portion of both acceptance and popularity. On the one hand, individuals tend to like their friends. But on the other hand, friendships are not strictly interpersonal, but tend to cluster in groups – and in groups, status competition usually plays a certain role. The liking may turn sour when the same individual in the friends’ group repeatedly interrupts you or others, tops your jokes by adding an even better pun, reveals secrets you shared with them, or gets physically domineering after the second beer.

In this assignment, you are asked to investigate how hierarchy can be found back in three networks: a friendship network, a network of popularity nominations, and a network of liking nominations. You will compare the same ERGM specification for these three networks separately, and then estimate another specification for the friendship network that takes information from the other two networks into account as predictors.

An important aspect of your investigation will be the question, *What determines a high indegree?* Merton (1968) explained the formation of hierarchies in collaboration networks by the so-called *Matthew effect*, a preference for collaboration with those who already successfully collaborate(d) with many others. The same phenomenon also holds for friendship networks, as sociometrists Moreno & Jennings (1938) already showed three decades before Merton. They called it by another name, *the sociodynamic effect*. Three decades after Merton, network scientists Barabási & Albert (1999) called the same thing *preferential attachment*, today the most commonly used term.

Exercises

Please answer the following conceptual questions (independent of any particular data):

- 1) Would you expect preferential attachment (or, equivalently, the Matthew effect) to be stronger for *liking* nominations or for *popularity* nominations? Or would you expect no difference at all? Explain your expectation!

- 2) What about friendship, when adding it to this comparison? Above I argued that friendship contains elements of both liking and popularity. If this is true, does that mean also the strength of the Matthew effect will then lie somewhere in the middle of the two other networks?
- 3) Argue why it is a good idea, when analysing empirical networks of adolescents, to include the network mechanisms of *gender homophily* (nominating same-gender persons), *reciprocation* (sending a nomination to those people one receives a nomination from) and *transitive closure* (closing directed two-paths by a direct tie in the same direction), even if the focus of the analysis is on hierarchy.
- 4) Please come up with an index which can express hierarchy and/or hub formation in a network, i.e., the differentiation of individuals in terms of their indegree. Justify your choice.

Now let us move to network modelling of empirical data, in which some of your reasoning from Exercises 1 and 2 can be quantitatively assessed, combining empirical analysis with simulation.

- 5) Evaluate the three network variables in your personalised data set in terms of density, reciprocity and transitivity, as well as gender homophily and your hierarchy index from Exercise 4. Put the results in a table next to each other to facilitate comparison.
- 6) Please fit an exponential random graph model to each of the networks, with the following specification:
 - overall tie creation tendencies (edges term),
 - reciprocation tendencies (mutual term),
 - gender homophily tendencies (*hint: use nodematch*),
 - transitive closure tendencies (*hint: use gwesp(0.7, fixed=TRUE)*)
 - the twopath-effect (as in the lab model scripts; this helps avoiding degeneracy-related estimation problems),
 - preferential attachment tendencies (*hint: use gwidegree(0.7, fixed=TRUE)*).

Report the converged results in a single table next to each other to facilitate comparison.

- 7) Please interpret these results in a brief text, in the light of your expectations formulated in Exercises 1, 2 and 3 above. Warning: These are complex generalized-linear models for three different data sets! Make sure that you only compare what can be compared!
- 8) Optional: Please also assess, report and discuss *effect sizes* for the preferential attachment effects found in the three analyses.
- 9) Give a brief assessment of *goodness of fit* of the three models.
- 10) Please fit another model but only to the friendship network. In addition to the above specification, this model shall also include main effects of acceptance and popularity nominations on friendship (*hint: use edgecov*).
- 11) Report converged results, and interpret by comparison to the simpler friendship model already estimated as part of Exercise 6. Does this bolster your reasoning in Exercise 2 above?
- 12) Please indicate whom you collaborated with when answering the above questions, and whose input you considered particularly helpful. *Explanation: this will not be counted against any of you. I realise that collaboration is a good thing when mastering complicated tasks. I ask for this network information only because it helps me giving feedback efficiently when correcting the assignments you will hand in.*

Consultation hour

After you have started working on the assignment in the lab class on February 13th, I (Christian Steglich) offer a max. 30 minutes online consultation before you submit your work. It will take place on **Thursday 19 February 10:00-10:30**, under this Zoom link:

<https://liu-se.zoom.us/j/68969751820?pwd=5Wmdcha1tEcRqz51N5GmxNfGSE5K5.1>

If you join without link, the Meeting ID is 689 6975 1820 and the Passcode is 654385.

Deadline

Please hand in your answers to the exercises by uploading an R markdown document to our course's Lisam environment: One is the R- or Rmd-script you used, the other is the MS Word or pdf document containing the answers to the exercises above. There are separate submission options for both. Note that by submitting online, there will routinely be a plagiarism check of the submitted documents. Deadline for handing in is **Friday 20 February 2026, 23:59**. The online submission option will close Monday morning.

Literature

Barabási, A. L., & Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439), 509-512.

Mayeux, Lara, Houser, John J., & Dyches, Karmon D. (2011). Social acceptance and popularity: Two distinct forms of peer status. In A. H. N. Cillessen, D. Schwartz, & L. Mayeux (Eds.), *Popularity in the peer system* (pp. 79–102). The Guilford Press. **Uploaded as a photocopy on Lisam in the "Readings" folder.**

Merton, Robert K. (1968). The Matthew effect in science: The reward and communication systems of science are considered. *Science*, 159(3810), 56-63.

Moreno, Jacob L., & Jennings, Helen H. (1938). Statistics of social configurations. *Sociometry*, 342-374.